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Gaze in the Crowd: Frustum-Aware Feature Aggregation for Robust Gaze Target Estimation

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 (https://www.catalyzex.com/add_code?title=Gaze in the Crowd: Frustum-Aware Feature Aggregation for Robust Gaze Target Estimation&paper_url=)

 21 Mar 2026 (modified: 09 Apr 2026)  ACMMM 2026 Conference Submission

 Conference, Senior Area Chairs, Area Chairs, Reviewers, Authors

 Revisions (/revisions?id=yCGuMGjDXZ)

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Abstract:

As a core capability for human intent inference, gaze target estimation remains challenging due to severe contextual interference in crowded scenes. Traditional multi-branch methods suffer from cumbersome architectures and heavy reliance on auxiliary modalities. Meanwhile, recent single-branch models based on Vision Foundation Models (VFM)s blindly exploit only the deepest features. Especially in crowded environments, this over-reliance on deep representations causes VFMs to struggle with semantic confusion and lose the high-resolution geometric details essential for accurate gaze tracking. To address this, we present GazeSpot, which unleashes multi-layer foundation features to restore the geometric-semantic balance via two key contributions: First, the Geometry-Guided Spatial Focus (GGSF) module leverages a physical frustum prior to generate an explicit multiplicative mask. Acting like a spotlight, GGSF proactively filters background noise and mitigates contextual interference to resolve spatial focus. Second, the Scale-Aware Semantic Aggregation (SASA) module adaptively fuses shallow, occlusion-resistant textures with deep semantics based on the global context, effectively addressing the feature granularity selection problem. We curate a specialized Crowd evaluation subset to facilitate robust benchmarking under severe social interference. Extensive experiments on general public benchmarks and the Crowd subset demonstrate that our method achieves state-of-the-art performance. Code and pre-trained models will be released.

Keywords: Gaze Target Estimation, Vision Foundation Models, Spatial Attention, Multi-scale Feature Aggregation, Human Behavior Analysis

Primary Subject Area: Multimedia Reasoning

Secondary Subject Area: Multimedia Interpretability and Explainability, Emotional and Social Signals, Multimedia Generative and Foundation Models

Submission Guide: Yes

Open Discussion: Yes

Author Registration Confirmation: Yes

Reviewer Participation: Yes

Desk Rejection Policy: Yes

Supplementary Material:  zip (/attachment?id=yCGuMGjDXZ&name=supplementary_material)

Submission Number: 3106

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Official Review of Submission3106 by Reviewer uFer

Official Review by Reviewer uFer 12 May 2026, 22:47 (modified: 28 May 2026, 21:44)
Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Reviewer uFer, Authors
Revisions (/revisions?id=WFOGEXaL90)

Strengths:

- Unlike methods that rely on single-layer VFM, GazeSpot effectively uses intermediate layers to restore the balance between fine-grained geometry and high-level semantics.
- The GGSF module's use of a "Spotlight Mask" provides a physically grounded way to filter out distractions, significantly improving interpretability and focus in multi-person scenes.
- The creation of a Crowd-specific evaluation trajectory allows for a rigorous quantitative study of how models degrade under increasing social interference.

Weaknesses:

- While the paper reports results on GazeFollow and VAT, the core claims regarding "robustness" are primarily validated on the authors' own Crowd subset. To truly demonstrate generalizability, the framework should be evaluated on other established benchmarks with diverse distributions, such as GOO-real.
- The model's reliance on a physical frustum prior makes it vulnerable to errors in head pose or bounding box estimation, as shown in the "Failure Cases" analysis. An analysis of robustness to noisy head boxes would strengthen the paper.

Review:

GazeSpot effectively addresses the "feature over-smoothing" issue in VFMs by leveraging multi-layer features. The GGSF module is a strong contribution. SASA further enhances this by dynamically weighting features across scales based on global context. The model achieves impressive SOTA results on GazeFollow and VideoAttentionTarget with a very lightweight learnable overhead. However, the paper could be strengthened by a more exhaustive comparison on OOD data, such as GOO, and by measuring robustness to annotation noise.

Fit: 4: Large audience

Fit Justification:

The paper aligns with human-centric multimedia analysis and scene understanding, and previous literature on this topic has been published in the same venue, making the submission a good fit for the conference.

Technical Quality: 4: Excellent

Technical Presentation: 5: Excellent

Rating: 4: Weak Accept

Confidence: 3: Knowledgeable

Add:

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Official Review of Submission3106 by Reviewer fs7N

Official Review by Reviewer fs7N 12 May 2026, 14:30 (modified: 28 May 2026, 21:44)
Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Reviewer fs7N, Authors
Revisions (/revisions?id=Dxb7D71F0A)

Strengths:

The proposed GazeSpot method has several prominent strengths for robust gaze target estimation in crowded scenes.

First, it explicitly addresses the spatial-semantic dilemma faced by single-branch vision foundation models, by effectively retaining high-resolution geometric details while maintaining powerful global semantic representations.

Second, the geometry-guided spatial focus (GGSF) module introduces an explicit physical frustum prior, which acts as a spotlight mask to suppress background interference and prevent attention hijacking, greatly improving spatial alignment and interpretability.

Third, the scale-aware semantic aggregation (SASA) module enables adaptive multi-scale feature routing, dynamically combining shallow occlusion-resistant textures and deep semantics according to the scene context.

Fourth, the entire framework remains highly efficient and lightweight with only a small number of trainable parameters, while achieving state-of-the-art performance on both standard benchmarks and the newly constructed crowded evaluation subset.

These strengths make GazeSpot particularly suitable for gaze target estimation tasks in real-world crowded, occluded, and highly interactive scenarios.

Weaknesses:

Despite its competitive performance, the proposed GazeSpot method still has several noticeable weaknesses.

First, the geometric prior constructed by the GGSF module is purely based on 2D position information, without considering the actual gaze direction, leading to limited adaptability when the head orientation and viewing direction are inconsistent.

Second, the geometry-guided spatial focus module highly relies on the quality of head bounding box priors. In cases of heavy occlusion, extreme head poses, or inaccurate detection, the generated frustum mask will become unreliable and degrade gaze prediction accuracy.

Third, the performance gain brought by our method is more significant in crowded scenarios but relatively limited in simple or sparse scenes, showing an uneven performance boost across different environments.

Review:

This paper addresses a practical and challenging issue in gaze target estimation, namely, performance degradation caused by contextual interference and feature over-smoothing in crowded scenes. The research motivation is clear and well-founded, with convincing observations on the spatial-semantic dilemma of single-layer foundation model features. The overall architecture of GazeSpot is well-designed: the GGSF module effectively suppresses background distractions using explicit geometric priors, while the SASA module reasonably achieves adaptive multi-scale feature fusion. Both components are lightweight and compatible with frozen vision foundation models, making the pipeline efficient and easy to reproduce. The experimental validation is comprehensive, including comparisons with state-of-the-art methods, detailed ablation studies, and a specially constructed crowd evaluation subset that strongly demonstrates the advantages in complex scenarios. However, the method still has certain limitations. The geometric prior relies too much on head detection quality, and the multi-layer fusion strategy lacks task-aware dynamic adjustment. Although the performance is improved significantly in crowded scenes, the enhancement in general scenarios is relatively marginal. Overall, this work is solid, innovative, and experimentally sufficient, making a valuable contribution to robust gaze target estimation in crowded environments.

Fit: 4: Large audience

Fit Justification:

This work targets gaze target estimation in crowded scenes, a highly practical and widely concerned topic in human behavior analysis, human-computer interaction, and visual intelligence. The proposed GazeSpot framework addresses the common spatial-semantic dilemma in vision foundation models, which is not only valuable for gaze estimation but also provides enlightening ideas for other dense prediction tasks with similar challenges. Moreover, the constructed crowd evaluation subset offers a new benchmark for robust model testing. Due to its practical significance, universal design philosophy, and extensible methodology, this paper can attract a large audience from multiple research communities including computer vision, robotics, and human-centered computing.

Technical Quality: 4: Excellent

Technical Presentation: 4: Good

Rating: 4: Weak Accept**Confidence:** 3: KnowledgeableAdd: **Official Comment****Official Review of Submission3106 by Reviewer epmb**

Official Review by Reviewer epmb 📅 11 May 2026, 20:17 (modified: 28 May 2026, 21:44)

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Reviewer epmb, Authors

📄 Revisions (/revisions?id=7pA8KkQdfO)

Strengths:

Deep and insightful motivation: The paper accurately identifies the "spatial-semantic dilemma" and the issue of deep feature over-smoothing faced by current single-branch Vision Foundation Models (VFMs, e.g., DINOv3) in dense prediction tasks. The PCA visualization comparing the last-layer features of DINOv2 and DINOv3 strongly supports this hypothesis.

Clever and intuitive method design: The proposed Geometry-Guided Spatial Focus (GGSF) module ingeniously utilizes the head bounding box as a physical prior to generate a spotlight-like frustum mask, actively and effectively filtering out background noise in crowded scenes. Concurrently, the Scale-Aware Semantic Aggregation (SASA) module achieves adaptive fusion of cross-scale features (shallow, mid, deep) via dynamic routing, elegantly balancing geometric details with high-level semantics.

Solid and targeted experiments: The authors not only achieve SOTA performance on mainstream benchmarks like GazeFollow and VideoAttentionTarget (VAT), but also construct a specific "Crowd" evaluation subset tailored to the paper's core problem. The analysis of the performance degradation trajectory under varying crowd densities is highly convincing.

Weaknesses:

Over-reliance on prior cues: As acknowledged by the authors in the "Limitation" section, the physical frustum isolation of the GGSF module heavily relies on the accuracy and quality of the target's head bounding box. When the person's head is severely occluded or at an extreme viewpoint (as shown in Figure 5), this spatial prior becomes unreliable, leading to attention diffusion. This may affect the model's robustness during end-to-end real-world deployment (e.g., when paired with an imperfect head detector).

Lack of inference latency evaluation: Although the paper emphasizes the extremely low number of learnable parameters (3.5M), extracting multi-layer features from a VFM and performing dynamic aggregation will inevitably introduce additional computational overhead compared to minimalist baseline models using only single-layer features (e.g., Gaze-LLE). The paper lacks practical deployment-related metrics such as FPS or inference latency to fully substantiate the efficiency claims.

Review:

The paper proposes GazeSpot, a highly efficient single-branch framework for robust gaze target estimation, particularly focusing on mitigating contextual interference in crowded social scenes. The authors highlight a critical bottleneck in modern Vision Foundation Models (VFMs): relying solely on the deepest layer leads to feature over-smoothing and a "spatial-semantic dilemma" where high-resolution geometric details are lost.

To overcome this, GazeSpot introduces two elegant modules. First, the Geometry-Guided Spatial Focus (GGSF) explicitly projects a physical "spotlight" mask derived from head coordinates to act as a physical shield against contextual interference. Second, the Scale-Aware Semantic Aggregation (SASA) module adaptively routes features across multiple layers, dynamically mixing shallow geometric textures with deep global semantics based on the global context.

The experimental validation is extensive. The authors evaluated the model on GazeFollow and VideoAttentionTarget (VAT), achieving SOTA results. More importantly, the curation of a specific "Crowd Subset" and the fine-grained degradation analysis perfectly prove the effectiveness of the proposed GGSF and SASA modules in complex social environments.

Overall, the paper is well-written, clearly motivated, and technically sound. The reason for a "Weak Accept" rather than a "Strong Accept" stems from the inherent limitation of the method: its heavy reliance on the upstream proxy (head bounding box quality). As shown in the failure cases, poor head detection in extreme occlusions

breaks the spatial prior. Additionally, while the parameter count is low, the actual inference latency (FPS) compared to single-layer models should be benchmarked. Addressing these points in the rebuttal or final version would strengthen the manuscript.

Fit: 4: Large audience

Fit Justification:

Gaze target estimation is a fundamental task for visual behavior understanding, human intent inference, and human-computer interaction (HCI). These areas are central themes within the ACM Multimedia community, making this paper highly relevant to a large audience interested in human-centric multimedia analysis.

Technical Quality: 3: Good

Technical Presentation: 4: Good

Rating: 4: Weak Accept

Confidence: 2: Familiar

Add: **Official Comment**

Official Review of Submission3106 by Reviewer CoU9

Official Review by Reviewer CoU9 📅 07 May 2026, 02:00 (modified: 28 May 2026, 21:44)

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Reviewer CoU9, Authors

📄 Revisions (/revisions?id=JVsMtyGKsF)

Strengths:

1. The proposed method achieves new state-of-the-art performance on public datasets.
2. This paper provides a feature analysis of different backbones, which helps in understanding the motivation.
3. The paper is well organized and easy to read.

Weaknesses:

1. The novelty of the proposed method is incremental and limited. The scale-aware semantic aggregation module is similar to classic multi-scale fusion methods.
2. The motivation and effectiveness of the geometry-guided spatial focus module are unclear. How does this module learn the spotlight without supervision?
3. ChildPlay and GOO-Real are also benchmark datasets in this domain; however, the paper does not evaluate the proposed method on these datasets.
4. There are some formatting errors, such as missing commas or periods at the end of equations.
5. The parameters in Table 2 may be misleading to readers. In fact, some values are learnable parameters rather than model parameters. The authors should fix this issue.

Review:

This paper focuses on the task of gaze target estimation. To address the issue of losing geometric details caused by overreliance on deep features in previous work, this paper proposes a new framework named GazeSpot that aims to leverage multi-layer foundation features to restore the geometric-semantic balance. Specifically, GazeSpot first employs a geometry-guided spatial focus module to generate an explicit multiplicative mask via a physical frustum prior. Then, a scale-aware semantic aggregation module is proposed to fuse textures with semantics, addressing the feature granularity selection problem. Extensive experiments are conducted to validate the effectiveness of the proposed method.

Fit: 3: Relevant to part of the community

Fit Justification:

The method shows strong performance, but the novelty is limited, and key components are not sufficiently justified. In addition, this paper with incomplete experimental evaluation. Therefore, I lean towards a weak reject.

Technical Quality: 2: Medium

Technical Presentation: 3: Fair

Rating: 2: Weak Reject

Confidence: 3: Knowledgeable

Add: **Official Comment**

Official Review of Submission3106 by Reviewer 7pTL

Official Review by Reviewer 7pTL 📅 25 Apr 2026, 17:10 (modified: 28 May 2026, 21:44)

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Reviewer 7pTL, Authors

📄 Revisions (/revisions?id=RGFXdhq6yN)

Strengths:

The paper tackles an important and relevant problem: gaze target estimation in crowded scenes, where contextual interference and multi-person ambiguity are practically significant. The proposed framework, GazeSpot, is conceptually simple: it uses a frozen vision foundation model, extracts multi-layer features, applies a geometry-guided spatial mask, and then performs scale-aware feature aggregation before heatmap decoding. The experimental section covers standard gaze-following benchmarks, including GazeFollow and VideoAttentionTarget, and additionally introduces a crowd-density evaluation subset. The reported results show some improvements over recent baselines, especially in L2 error on crowded samples.

Weaknesses:

1. The claimed “physical frustum prior” is not fully justified. According to the method description, the GGSF mask is generated from relative coordinates to the head center and the head box width/height, followed by a small MLP and sigmoid gating. This is closer to a learned coordinate-conditioned spatial attention mask than a physically meaningful gaze frustum, since it does not explicitly use head pose, eye orientation, depth, camera geometry, or a predicted gaze vector. The paper repeatedly uses terms such as “frustum,” “gaze cone,” and “physical isolation,” but the actual formulation appears substantially weaker. Prior works have already explored line-of-sight reasoning, gaze cones, geometric gaze fields, object-aware filtering, and multi-person gaze modeling, so the paper needs stronger comparisons and clearer positioning.
2. The central claim about the “spatial-semantic dilemma” and DINOv3 feature over-smoothing is not sufficiently supported. The evidence mainly consists of qualitative PCA visualizations and small performance differences. In Table 5, the DINOv3 last-layer baseline is worse than DINOv2 on GazeFollow and Crowd, but it is actually slightly better on VideoAttentionTarget L2. The differences are also quite small. This does not convincingly establish a general or severe over-smoothing phenomenon. Since DINOv3 itself was designed to improve dense visual representations, the paper should provide quantitative feature analyses, such as boundary separability, token similarity statistics in crowded regions, layer-wise probing, and statistical significance over multiple seeds.
3. The empirical gains are modest and sometimes inconsistent. On the public benchmarks, the strongest GazeSpot model improves over Gaze-LLE ViT-L, but the margins are relatively small. On VideoAttentionTarget, the base GazeSpot model has lower In/Out AP than several prior VFM-based baselines. On the Crowd subset, GazeSpot improves L2 in dense settings, but the In/Out AP is worse than Gaze-LLE at density 3 and density 4. These mixed results weaken the claim of robust superiority in crowded scenes. The paper would be much stronger with confidence intervals, repeated runs, significance tests, and a more complete comparison to multi-person gaze methods such as Sharingan and object-aware gaze-target methods.
4. The Crowd subset is under-specified. The paper defines crowd density by the number of annotated head bounding boxes and reports sample counts, but it does not provide sufficient details about the exact split, sample IDs, construction script, whether the subset will be released, or whether the distribution introduces confounding factors such as object category, scene type, annotation difficulty, or out-of-frame ratio. Since the Crowd subset is central to the paper’s contribution, its reproducibility and validity are essential.
5. The ablation study is incomplete. The paper compares last-layer features, raw multi-layer concatenation, SASA, and SASA+GGSF, but it does not compare against simple and strong alternatives: fixed Gaussian masks around the head, CoordConv-style coordinate channels, hand-designed gaze-cone masks, FPN-style feature fusion, layer-wise linear probing, different selected layers, different mask placements, or robustness to noisy head boxes. Without these controls, it is difficult to attribute the improvement specifically to the proposed “frustum-aware” design rather than to generic coordinate conditioning or multi-scale fusion.

Review:

This paper proposes GazeSpot, a VFM-based gaze target estimation framework for crowded scenes. The main idea is to extract multi-layer features from a frozen foundation encoder, apply a geometry-guided spatial focus module, and use scale-aware semantic aggregation before decoding the final gaze heatmap and in/out prediction. The problem is important and the proposed direction is reasonable. I appreciate the attempt to move beyond using only last-layer foundation features and to explicitly study robustness under crowd interference. The paper is readable and the experiments suggest that multi-layer feature aggregation can help gaze target estimation.

However, I lean toward weak reject because the current evidence does not fully support the paper’s main claims. The proposed GGSF module is presented as a physically meaningful frustum prior, but the actual implementation is a learned mask from head-box-relative coordinates. This makes the novelty less clear, especially given prior work on gaze cones, gaze fields, object-aware gaze modeling, and multi-person gaze following.

Fit: 4: Large audience

Fit Justification:

The topic is well aligned with ACM Multimedia, especially multimedia reasoning, human behavior analysis, social signal understanding, gaze estimation, and vision foundation models. The problem of gaze target estimation in crowded visual scenes is relevant to several ACM MM communities, including scene understanding, human-centered multimedia, HCI, and social behavior analysis.

Technical Quality: 2: Medium

Technical Presentation: 3: Fair

Rating: 2: Weak Reject

Confidence: 3: Knowledgeable

Add: [Official Comment](#)

Author Enrollment by Bin Fan

Author Enrollment by Bin Fan (/profile?id=~Bin_Fan6)  02 Apr 2026, 20:04  Conference, Bin Fan

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